

Ranking 13,891 Skills:

A Cross-Encoder Cascade and a Pre-Registered Measurement Protocol for Graded Skill Extraction

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Abstract

The RecSys-HR 2026 WorkRB challenge asks for a ranking of the entire ESCO skill taxonomy (13,891 concepts) by graded relevance for every query, across five datasets, scored by macro nDCG@100. This paper describes the entry that finished first, at nDCG@100 = 0.8458.

The system is a retrieve-then-rerank cascade: a two-lane fusion into a 1,000-candidate pool over which twelve retrieval lanes feed 33 per-candidate features, an architecture-diverse ensemble of listwise-trained cross-encoders (DeBERTa-v3-large and Qwen3-Reranker-4B/8B under LoRA), and a LightGBM LambdaRank fusion carrying a smoothed per-skill relevance prior and an exact-match `altLabel` override. None of those components is novel.

What decided the outcome was not the architecture but the protocol governing which components were allowed in. With 286 labelled validation queries and ten lifetime submissions, the binding constraint was *selection noise*, not model capacity: the smallest interesting improvement was an order of magnitude below the standard error of a single-task estimate. We describe the protocol that answered this: configuration lists pre-registered before each run, ten-seed paired comparison on identical folds, a strict conjunctive adoption gate, a fresh-seed confirmation as an explicit multiplicity guard, a winner’s-curse haircut sized to selection breadth, and a “slot bar” converting local cross-validation into a probability of improving on a given leaderboard row through a *t*-distributed prediction interval on the measured local-to-board offset. Under this protocol, across every configuration pre-registered and measured in the final competition phase, exactly one passed. We report all of them, passes and failures alike, with their numbers, the failures being the part of this work most likely to save someone else time.

Finally, we document a label channel that was openly reachable around the competition: the graded datasets’ test `qrels` splits were served, unauthenticated, through a public dataset API. We then report a *negative* result on the obvious follow-up question: whether a public leaderboard can reveal that such a channel was used. Two candidate statistics are described and both are withdrawn, one because it fires on a majority of the population and one because its apparent signal is confounded by the most ordinary act in competitive machine learning. No participant is named, and the operational thresholds are deliberately withheld.

1 Introduction

Skill extraction is the problem of reading a piece of text about work (a job advertisement, a course description, a résumé line) and identifying the skills it refers to, expressed against a controlled vocabulary. It underpins labour-market analytics, job matching, and curriculum design, and it is unusual among information retrieval problems in that the target space is fixed, enumerable, and hierarchical: the European Skills, Competences and Occupations taxonomy (ESCO) [2] v1.1 contains 13,891 skill concepts, each with a preferred label, a set of alternative labels, and a paragraph of description.

The RecSys-HR 2026 WorkRB challenge [1]¹ made two choices that separate it from earlier skill-extraction benchmarks [4, 8, 9, 13] and that shape everything below.

Graded, not binary, relevance. Every (query, skill) pair carries an integer grade in $\{0, 1, 2, 3, 4\}$: 4 for a skill explicitly demonstrated by the text, 3 for one strongly implied, 2 for a taxonomic neighbour at the wrong granularity, 1 for a plausible in-domain skill, and 0 for everything else. A system that pushes near-miss neighbours hard to zero (the natural behaviour of a model trained on binary labels) forfeits the partial credit at grades 1 and 2 while gaining nothing. This single design decision invalidates the most obvious training recipe, and we measured it doing so (§5).

Ranking the whole taxonomy. The system must score all 13,891 concepts per query, not select from a shortlist. Combined with nDCG@100, this makes the problem a cascade problem [36]: recall into a candidate pool and ordering within it are separable concerns with very different cost profiles.

Five datasets are scored and averaged with equal weight: four *skill-extraction* sets (House, Tech [4, 5], TechWolf [6], SkillScape [7]), whose queries are job-posting sentences or segments, and one *skill-normalisation* set (SkillNorm), whose queries are short surface forms to be mapped onto canonical ESCO entries. The organisers report 63 participants and 322 submissions; the public leaderboard at close carried 98 rows from 29 accounts.

1.1 Contributions

1. A complete description of the winning system (§3), including the two largest gains that required no GPU at all: a smoothed per-skill relevance prior (+0.0380 local, verified over seven disjoint seeds, a leave-one-task-out control and a label-shuffled control) and an exact-match `altLabel` override (+0.0564 on the normalisation task locally, exactly 0.0000 on the other three, projecting to +0.0113 on the five-task average).
2. A measurement protocol for competitions where the validation set is too small to support both selection and honest estimation (§4). We argue this is the transferable part of the work.
3. A full ablation record (§5), reported as a ledger of outcomes rather than a highlight reel: every configuration measured in the final phase, of which one was adopted.
4. A description of an openly reachable label channel, and a negative result on leaderboard forensics (§6): two plausible detection statistics, both withdrawn, with the base-rate checks that killed them. Stated without reference to any participant.

1.2 Related work

Skill extraction has been posed as sequence labelling over job postings (SkillSpan [4], Kompetenzer [8], and the benchmark corpus of Green et al. [9]), and, once the target is a fixed taxonomy, as extreme multi-label classification or retrieval against ESCO [5, 6, 14]. Senger et al. [13] survey the area. Normalisation, mapping a surface form onto a canonical entry, has been treated as entity linking [11] and supported by taxonomy-aware pre-training [10]; the limits of prompting alone were examined by Nguyen et al. [12]. Synthetic postings have been used to widen coverage [7]. WorkRB [1] unifies these strands into one benchmark, and this paper describes an entry to its first challenge.

Two features of that challenge push it away from the prior formulations. Graded relevance makes the target an ordering rather than a set, so the natural objective is a listwise ranking loss [40, 42] rather than a classification loss; and scoring the whole taxonomy per query makes

¹Hosted on Codabench [3], competition 16900. Challenge repository: <https://github.com/techwolf-ai/workrb-recsys-hr-challenge-2026>.

Dataset	Val queries	Test queries	Median len.	Kind
House	61	261	20 words	extraction
Tech	75	338	10 words	extraction
TechWolf	–	324	13 words	extraction (no local labels)
SkillSkape	100	500	27 words	extraction
SkillNorm	50	450	4 words	normalisation
Total	286	1,873		

Table 1: Split shapes, measured by instantiating the WorkRB [1] tasks. All splits are English. The target space is the same 13,891 ESCO v1.1 concepts for every task.

recall and ordering separable, which is the classical argument for a cascade [36]. The architecture that follows is therefore standard retrieve-then-rerank [37, 38] rather than anything specific to this domain. What is not standard, and what this paper is mostly about, is the protocol governing which components were allowed into it.

1.3 A note on the final standing

At the close of the test phase the public leaderboard carried 98 rows from 29 accounts, and this entry’s best row ranked tenth on score. The organisers then verified rule compliance across entries, a process they described as taking considerably longer than expected, and the final standing announced afterwards placed this entry first. I have no visibility into that process. This paper does not speculate about the compliance status of any other entry; §6 discusses only a data-availability fact and a negative result about leaderboard statistics.

2 Task, data, and metric

2.1 The five datasets

Table 1 gives the shape of the problem. Two features of it drive most of the design.

First, **TechWolf has no validation split at all**. It is one fifth of the final score and 324 of the 1,873 test queries, and no local signal about it exists. Any decision made to help TechWolf is a generalisation bet placed blind. We measured that it is structurally an extraction task rather than a normalisation one: its `altLabel` exact-match hit rate is $3/324 = 0.9\%$, against $450/450 = 100\%$ for SkillNorm. It was treated as Tech/House-like throughout.

Second, **the validation set is small**: 286 queries in total, and SkillNorm’s 50 queries carry roughly ± 0.03 of noise on their own. The paired minimum detectable effect at 80% power on $n = 50$ is ± 0.015 to ± 0.017 . The improvements that actually separated the top of the leaderboard were between $+0.001$ and $+0.01$. That mismatch is the central methodological problem of this competition, and §4 is the response to it.

2.2 The metric, and one trap inside it

Scoring is macro-averaged nDCG@100 [31] with exponential gain:

$$\text{DCG@100} = \sum_{i \leq 100} \frac{2^{\text{rel}_i} - 1}{\log_2(i + 2)}, \quad \text{nDCG@100} = \frac{\text{DCG@100}}{\text{IDCG@100}}, \quad (1)$$

with the ideal ranking taken over all annotated grades and truncated at 100, and all unannotated pairs treated as grade 0. The overall score is the *flat* mean over the five (task, `en`) datasets.

Two consequences are worth stating because both cost us time.

The metric is order-only. Any strictly monotonic transform of the scores leaves nDCG@100 unchanged. Calibration is therefore a no-op for a single final ranker; it matters only when combining lanes whose scores must be made commensurable, which is why every lane below enters as a per-query z -score rather than a raw similarity.

The obvious local aggregation is the wrong one. WorkRB’s `get_summary_metrics()` aggregates hierarchically by task group. SkillNorm is its own group (SKILL__NORMALIZATION) while the four extraction tasks share one (SKILL__EXTRACTION), so the default summary weights SkillNorm at roughly 50% instead of the flat 20% the leaderboard uses. Optimising the default number optimises the wrong objective. I computed the flat mean explicitly and later verified the leaderboard does the same, across all nine of my own placed rows, to within 0.0001. That is what four-decimal display rounding permits; the largest single discrepancy was 0.00006.

2.3 Grades, and why binary training fails

The grade scale is semantic, not confidence-based:

Grade	Meaning
4	correct; explicitly demonstrated by the text
3	strongly relevant; implied
2	adjacent; right area, wrong granularity (an ESCO parent or child)
1	plausible; in-domain but not asked for
0	unrelated, and every unannotated pair

Validation **qrels** are *dense*: essentially every one of the 13,891 targets is graded for every query. Measured averages per query are 1.5–4.6 golds at grade 4, 2.6–10.0 at grade ≥ 3 , and 546–578 at grade ≥ 1 .

That density has a diagnostic consequence that misled us early. Because 546–578 targets per query carry grade ≥ 1 , recall of grade- ≥ 1 items into a top-200 pool is mathematically capped near 35%, and reads as catastrophic when it is not. The meaningful gate is recall of grade- ≥ 3 items into the top 100, which our fused retrieval reached at 0.654–0.800 by task, with grade-4 recall@100 at 0.842–0.960. Candidates were never the bottleneck; ordering was.

It also explains why binary training targets fail. Grades 1 and 2 are real credit, awarded precisely for surfacing taxonomic neighbours, but they must rank *below* the exact and implied items. A cross-encoder trained to drive neighbours to a hard zero destroys that ordering. We measured this directly: a MiniLM-L6 cross-encoder [20] fine-tuned on binary targets scored -0.0758 against the base it was meant to improve.

3 The system

The pipeline is a four-stage cascade: retrieve, featurise, rerank, fuse.

3.1 Stage 1: retrieval and the candidate pool

Twelve retrieval lanes score every ESCO concept for every query:

Family	Lanes
Domain bi-encoders	CurriculumMatch, ConTeXTMatch, jobbert, curricdesc
General bi-encoders	minilm, minilmdesc
Lexical	tfidfword, tfidfchar, bm25 [32], bm25desc
String	editdist, fuzzset

Backbones are Aleksandr/ skillmatch-mpnet-curriculum-retriever (CurriculumMatch), TechWolf/ConTeXT-Skill-Extraction-base, TechWolf/JobBERT-v2, and all-MiniLM-L6-v2: respectively a curriculum-retrieval bi-encoder [18, 17, 21], the ConTeXT skill-extraction encoder and JobBERT-v2 [16, 15], and a general-purpose sentence encoder [19, 20]. Lanes suffixed `desc` score against "{title}. {description}" rather than the bare ESCO title; `tfidfchar` uses character n -grams (`analyzer="char_wb"`, $n \in [1, 3]$).

The candidate pool comes from a two-lane z -weighted fusion [33, 34],

$$s_{\text{base}} = 0.6 \cdot z(\text{CurriculumMatch}) + 0.4 \cdot z(\text{ConTeXTMatch}), \quad (2)$$

taking the top $K = 1000$ concepts per query. Reranking depth is monotonic in this problem: measured on a fixed reranker with identical folds and features, top-100 gives 0.6852, top-200 gives 0.7226, and top-500 gives 0.7393. That ladder is what motivated exporting at $K = 1000$, and the later export confirmed the extrapolation: depth 1000 beat depth 500 by +0.0029 on lanes plus the cross-encoder alone, and by +0.0073 once the relevance prior and the `altLabel` override were added. Candidates and features are therefore exported at $K = 1000$, from which any shallower depth can be recovered for free. The emitted submission keeps the top 500 per query, deeper rows being a no-op at `nDCG@100`.

Two further details matter. First, this fusion was chosen by measurement, not intuition: naive equal-weight reciprocal-rank fusion [35] of the three obvious lanes *hurt*, scoring 0.5935 against CurriculumMatch’s 0.6416 alone. Second, the fused base at 0.6499 is the number every later stage is measured against, and reproducing it exactly served as a known-value gate that aborts the run when the pipeline drifts.

3.2 Stage 2: 33 features per candidate

Each of the 12 lanes contributes two columns gathered at the candidate positions: the per-query z -score, and $\log(1 + r)$ of the within-query rank. Nine aggregate columns follow: `base_z`, `base_logrank`, `agree50` and `agree200` (how many lanes place the candidate inside their own top 50 / 200), `zmean`, `zstd`, `zmax` across the lane block, and the word counts `target_len` and `query_len`. Total: $2 \times 12 + 9 = 33$.

The agreement and dispersion columns are the interesting ones: they let the fusion model learn that a candidate which every lane likes moderately differs from one that a single lane loves.

A finding worth isolating. **Every lane other than the two base retrievers is weak on its own**, scoring 0.095 to 0.53 macro `nDCG@100` for the nine with a measured macro, against CurriculumMatch’s 0.6416. All ten were cut as rankers. As *features* they are a different matter, and the description-scoring lane `curricdesc` ranked top-4 in feature importance despite scoring worse standalone than the plain CurriculumMatch lane it was derived from. Standalone quality is a poor guide to whether a signal belongs in a fusion; what matters is what the signal knows that the others do not. The same principle reappears at the reranker level, where it decided the final ensemble.

3.3 Stage 3: the cross-encoder ensemble

The reranker stage is where the compute went. Each member is a cross-encoder [22, 23, 38] trained in two stages and exported as out-of-fold validation scores plus test scores.

Two-stage training. Stage 1 pre-trains on a subsample of TechWolf/Synthetic-ESCO-skill-sentences (138,260 rows available, CC-BY-4.0; 80,000 sampled for the `bge-reranker-base` [26], DeBERTa-v3 [24, 25] and Qwen [28, 29] members, 30,000 for `bge-reranker-v2-m3` [27]), mixed with up to 14,000 ESCO description-to-skill rows that

supply the short-phrase supervision the synthetic sentences do not contain, for roughly 94,000 listwise rows in total. The grade ladder is synthetic, built by mining hard negatives with the ConTeXT bi-encoder. Each listwise row holds fifteen documents: one gold at grade 4, three near-synonyms at 2, five plausible mid-pool items at 1, and six deep random negatives at 0. Stage 2 fine-tunes on the real graded validation `qrels` inside the candidate pool, in three query-grouped folds.

The loss. All members train under LambdaLoss [42] with the NDCGLoss2++ scheme ($\mu = 10$), a listwise objective that weights each pair by the nDCG@100 change swapping them would cause, in the lineage of RankNet and LambdaRank [39, 40]. This is the natural objective when the metric is nDCG@100 and the labels are graded, and it is the direct reason binary targets were never an option after the first measurement.

One implementation note, recorded partly as a correction. The training code carries a guard dropping queries with no grade- ≥ 2 item inside the candidate set, on the reasoning that a listwise row with all-identical labels has no defined ideal ranking and is a live NaN source under LambdaLoss, and a comment in that code attributes four diverged local runs to it. That attribution turned out to be wrong. A later direct measurement refuted the degenerate-row hypothesis outright: zero of 1,144 and zero of 2,860 sampled rows had all-identical labels, so the ideal DCG was never zero. The divergences tracked step count instead, and their root cause was never established. The guard stays in as defensive coding, but it did not fix what the comment says it fixed. I leave the note in because a plausible root cause written confidently into a code comment is exactly the kind of thing that survives into a paper unchecked.

Members. The ensemble spans two architecture families deliberately:

Key	Backbone	Adaptation	OOF nDCG@100 alone
A	BAAI/bge-reranker-base (278M)	full	0.7327
A2	bge-reranker-base, denser stage 2	full	0.7490
A4	bge-reranker-base, 160×48	full	0.7666
B2	BAAI/bge-reranker-v2-m3 (568M)	full	0.7955
C2	microsoft/deberta-v3-large, 40×32	full	0.8161
C3	DeBERTa-v3-large, 80×32	full	0.8212
C4	DeBERTa-v3-large, 160×32	full	0.8245
Q	Qwen/Qwen3-Reranker-4B	LoRA [30]	0.8334
Q2	Qwen3-Reranker-4B, 40×32	LoRA	0.8375
Q8	Qwen/Qwen3-Reranker-8B, 16×24	LoRA	0.8358
Q8D	Qwen3-Reranker-8B, 40×32	LoRA	0.8407

($n \times m$ denotes the stage-2 density: n listwise rows sampled per query, m documents per row. Q8’s figure is derived rather than logged directly: the ledger records Q8D as +0.0049 over Q8 at the column level. Members in **bold** in later tables are the four that survived into the submitted ensemble.)

The last row deserves attention, because it is the clearest single illustration of what this competition rewards. **Q8D was the strongest individual reranker ever measured in this campaign**, and it is not in the final system. Every ensemble containing it failed the adoption gate (§5). Its skill was already present in the members it would have joined; what the ensemble was short of was not strength but *decorrelation*. The member that did pass, C4, was weaker standing alone than three of the columns already in the pool, and belonged to the *other* architecture family.

The converted head. Scoring Qwen3-Reranker through `AutoModelForSequenceClassification` attaches a randomly initialised classification head to what is architecturally a causal language model, discarding the reranking ability the checkpoint was trained

for. Instead we load `AutoModelForCausalLM` and score with the difference of two rows of the existing output embedding:

$$\text{score}(q, d) = \langle h_{-1}(q, d), W_{\text{lm}}[\text{yes}] - W_{\text{lm}}[\text{no}] \rangle, \quad (3)$$

where h_{-1} is the final hidden state at the last position. This identity was verified to 2.4×10^{-8} against evaluating the full logit vector, and it turns a 151k-way softmax into a single dot product. The resulting scalar head is then trained with LoRA ($r = 16$, $\alpha = 32$, dropout 0.05) on all attention and MLP projections.

An earlier attempt on a different checkpoint failed for exactly the reason this trick avoids: `mxbai-rerank-large-v2` routed through the generic cross-encoder path produced a random head and was cut.

3.4 Stage 4: fusion, and two levers that needed no GPU

The final ranker is a LightGBM [43] `LGBMRanker` with the `lambdarank` [41] objective, `label_gain` = [0, 1, 3, 7, 15] (the exponential gain of the metric) and `ndcg_eval_at` = [100], over the 33 retrieval features plus two columns per cross-encoder member (z -score and log-rank). It is trained with query-grouped 5-fold cross-validation over the 286 validation queries, consuming strictly out-of-fold cross-encoder scores, then refit on all queries as a bag of ten seeds.²

Two additional feature blocks carried more weight than any single reranker upgrade, and neither touched a GPU.

A smoothed per-skill relevance prior (+0.0380). Some ESCO concepts are simply relevant more often than others. Four columns capture that, each counted over the candidate pools of the *training* queries of a fold only, leave-one-out on training rows:

$$\frac{r+1}{s+4} (P[\text{grade} \geq 1]), \quad \frac{h+0.5}{s+8} (P[\text{grade} \geq 3]), \quad \log(1+s), \quad \frac{g+1}{s+4} (\mathbb{E}[2^y - 1]),$$

where s counts appearances in a pool, r and h count grade- ≥ 1 and grade- ≥ 3 occurrences, and g accumulates exponential gain. The effect was +0.0380 over seven disjoint seeds, survived leave-one-task-out at +0.0315, and collapsed under a label-shuffled control. The ladder in Table 3 credits the prior with +0.0331 rather than +0.0380; the two are different measurements, the first a rung of the v6 pipeline at candidate depth 1000 and the second an isolated lever averaged over seven seeds, and I quote each where it was taken. In the submitted model `prior_hi` is the highest-importance feature of all and `prior_egain` is fourth, behind the Q2 and Q8 log-rank columns; in the earlier single-cross-encoder configuration the two prior columns were the top two. Four columns of counting statistics outrank a four-billion-parameter reranker’s score.

An altLabel exact-match hard override. If a query string matches an ESCO alternative label exactly after normalisation, that concept is forced to the top at write time rather than offered as a feature. ESCO carries 97,441 alternative labels, 7.01 per concept on average, collapsing to 96,684 distinct normalised strings, of which only 0.74% are ambiguous (maximum fan-out 4).

Two separate measurements describe its effect, and they should not be confused. In isolation the override is worth +0.0564 on SkillNorm and exactly 0.0000 on each of the other three validation tasks, which averages to +0.0141 over the four local tasks and projects to +0.0113 over the five-task board. Measured as a rung of the pipeline ladder, with the relevance prior already in place, it moved the four-task macro by +0.0152; the two figures differ because the

²The fusion process imports only `numpy` and `lightgbm`. LightGBM segfaults without a traceback in any process where `torch` was imported first; not importing `torch` was the only fix that was not a workaround.

ladder rung includes the interaction with everything beneath it. It was never isolated on the leaderboard itself, since no two submissions differed only by this switch. The asymmetry is the whole story: 450 of 450 SkillNorm queries hit an exact `altLabel`, against 3 of 324 TechWolf queries.

3.5 Compute

Training ran on the NUS School of Computing compute cluster (A100-80GB, H100-96GB and H200-141GB nodes) under Slurm, for a campaign total of roughly 250–300 single-GPU hours. Two properties of the setup are worth noting for reproducibility: training length is derived from *measured throughput* against a wall-clock budget rather than a fixed epoch count, so the learning-rate schedule still decays to zero under a hard deadline; and a CUDA out-of-memory error is caught to halve the batch and double the step count, so the same number of rows is seen either way. DeBERTa-v3-large proved to have an architecture-specific memory profile that made dense configurations infeasible on both A100-80 and H100-96 (one run left 91GB of residue after two folds) and required H200-141 nodes.

4 The measurement protocol

This section is the part of the paper I would most like other participants to argue with, because it decided far more of the outcome than any architecture choice.

4.1 The problem it solves

Consider the position after the ensemble reached roughly 0.864 locally. The remaining candidate improvements were worth +0.001 to +0.002. The validation set is 286 queries. SkillNorm alone carries ± 0.03 of sampling noise. Dozens of plausible configurations remained unmeasured. Ten leaderboard submissions existed in total, and each returned exactly one number, hours later.

Under those conditions the default competition workflow (try things, keep what scores higher locally, submit the best) is not merely inefficient. It is a machine for manufacturing improvements that do not exist. The failure is well documented: selection on a finite validation set biases the selected model’s score upward [49], repeated adaptive querying of a held-out set erodes its validity [50, 51, 52], and measured across real competitions the resulting leaderboard gains are often smaller than they appear [53] or absent under fair comparison [54]. With enough correlated candidate configurations, something will always look better by +0.002.

4.2 Pre-registration

Every experimental round was written into an append-only ledger *before* the runs existed, enumerating the exact configurations to be tried and closing the space: *no configurations outside these templates without a fresh registration line written before the run*. Entries are appended, never rewritten; a correction is a new entry.

The purpose is not bookkeeping. It is to make the denominator of the multiple-comparisons problem knowable [57]. A protocol that permits configurations invented after seeing results has no denominator, and therefore no defensible p -value, no matter how it is analysed afterwards. This is the argument for pre-registration made in the experimental sciences [44, 45], and the degrees of freedom it closes are those of the garden of forking paths [46].

4.3 Paired comparison on identical folds

Fold assignment is deterministic in (seed, number of queries), so a per-seed baseline vector stored during an earlier round pairs exactly with a candidate run measured later. Every comparison

is therefore paired at the seed level over seeds 0–9, and the quantities reported are the paired mean difference, the number of seeds won, and the standard deviation across seeds. Pairing is what makes the comparison sensitive at this effect size; the caveat is that resampled runs share training data, so their differences are not independent and the naive variance across seeds understates the true uncertainty [47, 48].

Pairing validity was itself checked rather than assumed. Before the blend round, three configurations were required to reproduce the earlier harness seed for seed; parity was recorded as exact to 0.000000 on all three checks.

4.4 The adoption gate

A candidate is adopted only if all three conditions hold simultaneously against the in-run incumbent:

$$\Delta_{\text{paired}} > +0.0010, \quad \text{seeds won} \geq 8/10, \quad \text{sd} \leq 0.0015. \quad (4)$$

The strictness was enforced literally. In the final round, configuration **c4b** = (C2, C3, C4, Q2, Q8) returned $\Delta = +0.0010$ with all ten seeds won, and was recorded **FAIL**, on the strict inequality. The registration had pre-committed: *near-miss = fail, no exceptions, no post-hoc re-argument*. I mention it because that decision is what makes every other number in this paper worth reading. A gate that bends once is not a gate.

4.5 The multiplicity guard

Any configuration passing the gate faced a confirmation run on *fresh* seeds 10–19, keeping its pass only if it also beat the incumbent there ($\Delta > +0.0005$, $\geq 7/10$ seeds). This was exercised exactly once, on the only passer of the extended phase: **c4a** = (C2, C4, Q2, Q8) returned $+0.0010$ with 9/10 wins on the fresh folds. Its discovery estimate was 0.8647; the pooled twenty-seed estimate was ≈ 0.8644 . The direction is what the winner’s curse [56, 49] predicts and the size matches the 0.0003 haircut registered for it in advance, but against a measured across-seed standard deviation of 0.0007, a single 0.0003 gap is well inside noise. Read it as consistent with the curse, not as a measurement of it.

When selecting among many correlated subsets (one round evaluated 26) a haircut of 0.0003 was subtracted from the winner’s paired mean before comparing it to the decision threshold, sized to the selection breadth.

4.6 The slot bar: turning local CV into a probability

The remaining question is when a locally better configuration is worth one of ten irreplaceable submissions. This requires mapping local cross-validation onto the leaderboard, and the mapping is empirical.

Across the campaign the local-to-board offset was measured seven times, and it *moved as the pipeline changed*, which is the first thing to say about it, because it is the thing that makes naive use of it dangerous. The two zero-shot baselines read -0.0124 and -0.0139 ; the first fitted submission read -0.0086 ; and the mature pipeline settled between -0.0192 and -0.0235 . A fitted system transfers worse than a zero-shot one, and the offset roughly doubled as the system grew.

The first two measurements were nonetheless *free*: two zero-shot baselines already present on the public leaderboard could be scored locally, giving an offset estimate without spending a submission at all (CurriculumMatch $0.6416 \rightarrow 0.6292$; ConTeXTMatch $0.6214 \rightarrow 0.6075$). They were superseded once real rows existed, and the thresholds below rest on the four most recent own-row offsets only, which do concentrate: mean -0.02005 , standard deviation 0.00077.

Given a target board score b , the local score required to beat it with probability p follows from a one-sided prediction interval for a *new* offset draw. The quantity being predicted is a

future observation rather than a mean, which is also why the cross-validation estimate itself carries more uncertainty than its across-fold spread suggests [55]. With only four offsets in the final estimate, this is a t -interval on three degrees of freedom, not a normal z -interval. A critic caught the distinction, and it moved the thresholds materially. Using $\text{sd}_{\text{pred}} = 0.00086$:

Candidate family	local for $\approx 50\%$	for $\approx 90\%$
Blend family (offset centre -0.0208)	0.8654	0.8668
Lean single-config family (centre -0.0194)	0.8640	0.8654
Unconditional	0.8647	0.8661

These bars were registered before the final round. They are what made the last decision mechanical: the adopted configuration **c4a** sat at 0.8647 local, clearing the lean-family 50% bar of 0.8640 but not the 90% bar, and it was submitted on exactly that basis. It scored 0.8458 against the board row it needed to beat at 0.8444.

4.7 What the protocol cost

It is fair to record the price. The protocol rejected **c4b**, which beat its incumbent on ten seeds out of ten, the closest call of the campaign. It rejected three Round-Q2D configurations at $+0.0009$, $+0.0009$ and $+0.0006$ with 8–9 seeds won. It rejected an ESCO attribute join at $+0.0005$ with 6/10, the closest of the feature-block rejections. And it refused a per-task recombination of my own already-scored leaderboard rows, deterministically worth $+0.0002$ and not excluded by any rule I could point to, though my own review recorded the rules-letter status as ambiguous, since the rules require the submission file to be produced by the real benchmark tasks.

I believe every one of those was correct, and I would make the same calls again. But a protocol tuned this conservatively will, in expectation, leave small real gains on the table. The trade is deliberate: in a regime where the measurement error exceeds the effect size, the dominant risk is not missing a real gain; it is adopting a false one and building on top of it.

5 Results

5.1 Final result

	House	Tech	TechWolf	SkillScape	SkillNorm	Macro
Submitted system	0.8387	0.8390	0.8347	0.8683	0.8480	0.8458

Table 2: Final leaderboard result (test split, submission 874278). Configuration (C2, C4, Q2, Q8), candidate depth 1000, prior and **altLabel** override enabled, bagged over ten seeds. Build cross-validation over three seeds was $0.8649/0.8659/0.8652 \rightarrow 0.8653$, giving a realised local-to-board offset of -0.0195 , inside the pre-registered lean-family band of -0.0192 to -0.0199 .

Note the shape of that row. The submitted system is the only entry in the top nine whose SkillNorm score *exceeds* its own extraction average: 0.8480 against 0.8452. Two things produce that sign and only one of them is measured. The **altLabel** override lifts SkillNorm and nothing else. But this entry’s extraction scores also sat roughly 0.05 below the top of the board, which lowers the other side of the same comparison. The inversion says as much about where this system was weak as about where it was strong.

5.2 The ladder from baseline to final

Two observations, once the GPU column is read alongside the deltas.

Stage	What was added	Local macro	Δ	GPU?
Lexical baselines	BM25	0.2618	–	
	TF-IDF	0.3068		
Domain retrievers	ConTeXTMatch (zero-shot)	0.6214		
	CurriculumMatch (zero-shot)	0.6416		
Naive fusion	RRF(ConTeXT, Curric, BM25)	0.5935	−0.0481	
z -weighted fusion	$0.6 z(\text{Curric}) + 0.4 z(\text{ConTeXT})$	0.6499	+0.0083	
Learning to rank	9 lanes $\rightarrow$ LambdaRank	0.6684	+0.0185	no
	12 lanes	0.6790	+0.0291	no
+ graded cross-encoder	MiniLM CE as a feature	0.7178	+0.0388	yes
+ depth and capacity	bge-base at top-500 (v5)	0.7393	+0.0215	yes
	bge-base at depth 1000 (v6)	0.7549	+0.0156	yes
+ relevance prior	4 smoothed prior columns	0.7880	+0.0331	no
+ <code>altLabel</code> override	exact-match hard override	0.8031	+0.0152	no
Final ensemble	(C2, C4, Q2, Q8), 10-seed	0.8647	+0.0616	yes

Table 3: The ladder, in local validation terms. Deltas are against the row above, except within the zero-shot block. Note that naive equal-weight RRF is *worse* than its best single component. Comparisons across the middle of this table are approximate: the pipeline changed underneath them (candidate depth moved from 200 to 500 to 1000, and the fold protocol from 3 seeds to 10), so the rows describe the campaign’s actual trajectory rather than a controlled ablation.

First, the three interventions requiring **no GPU at all** (learning to rank over the retrieval lanes, the relevance prior, and the `altLabel` override) are together worth about +0.077 over the fused base, roughly a third of the total gain, at a rounding error of the cost. The prior alone (+0.0331) is worth more than fusing twelve retrievers with a learned ranker (+0.0291).

Second, the cross-encoder stack is nonetheless where most of the score is: roughly +0.138 across its four steps, and the single largest step in the table is the final ensemble at +0.0616. The expensive part is expensive because it works. What the table argues is not that the neural stage was unnecessary but that it was *mis-sequenced* in my own planning: the cheap levers were available from day one and I did not reach the prior until the campaign was two-thirds gone.

And at the far end, the entire extended phase (weeks of dense reranker training on cluster GPUs) moved the local score by +0.0011.

5.3 The final-phase ablation ledger

Table 4 records every configuration measured in the extended phase. Each was pre-registered before it ran, and each is reported here whether it passed or not.

5.4 Negative results

The following were measured and rejected. Each cost real time, and each is reported so that it need not cost anyone else’s.

Off-the-shelf rerankers are not a free win.

A zero-shot `bge-reranker-base` applied over the top-200 candidates of the best domain retriever *lost* −0.061 macro, and lost on every task: House −0.036, Tech −0.069, SkillScape −0.102, SkillNorm −0.038. Grade-4 recall inside the reranked set was 0.87–1.00, so this is an ordering failure, not a recall failure. A general-purpose web-passage reranker orders skill concepts worse than a domain-tuned cosine. Fine-tuning on the actual grade scale is what makes a cross-encoder useful here.

Binary training targets.

A MiniLM-L6 cross-encoder fine-tuned on binary relevance measured −0.0758. See §2.

Synthetic query generation.

Generating additional labelled queries with an LLM and mixing them into stage-2 training measured -0.0094 at full dose against a $+0.005$ pass threshold, and was killed. The root cause was a content mismatch rather than a quality problem: synthetic House queries were *too easy* (base nDCG@100 0.644 against the real 0.606), and House, which took the largest dose, lost the most (-0.016). The measured slope of real labelled data was $+0.01266$ per doubling; synthetic data did not sit on that curve.

The ESCO hierarchy.

The taxonomy’s graph structure is the obvious missing signal, and we could not get it. The documented bulk downloads returned 404 and 401; the public REST API does serve broader/narrower concepts, but a full pass is 13,891 requests and egress began failing after eight concurrent calls. Measured against the flat attribute features we did have, graph edges contributed $+0.0000$. The hierarchy half was cut outright.

ESCO item attributes.

Skill type, reuse level, $\log(1 + \text{\#altLabels})$ and $\log(1 + \text{description length})$, joined at 13,891/13,891 coverage, measured $+0.0005$ with 6/10 seed wins, the closest of the feature-block rejections, and still a failure by the gate.

A cross-encoder disagreement feature.

One standard-deviation column across the cross-encoder z -scores measured 0.8641, identical to baseline to four decimals. The fusion already sees disagreement implicitly through the individual member columns.

An altLabel substring feature.

Distinct from the exact-match override that worked: substring containment as a soft feature fired on 27–62 extraction queries and measured 0.8616 against a 0.8620 baseline. Killed.

A kNN graph lane over ESCO embeddings.

Cut.

Stripping bullet prefixes.

Leading " $*$ " and " $-$ " markers appear in 42/338 test Tech queries and 169/324 test TechWolf queries, and stripping them looks obviously correct. Measured on Tech in a smoke test over a subsampled corpus it *lost* -0.0120 ; at base level on the full corpus and all 75 validation Tech queries it lost again, -0.0028 on Tech and -0.0007 on the macro. The full-pipeline recheck at ensemble level was never run (it would have cost cluster hours against a single remaining submission slot) so I have no confirmed explanation, only the suspicion that the encoders learned something from list structure. Not applied.

LightGBM hyperparameter search.

A sweep selected settings that reversed on disjoint seeds. The sweep was abandoned and the parameters locked for the remainder of the campaign. This is selection noise in its purest form: a hyperparameter search over a 286-query validation set is mostly a search over the noise.

Instruction conditioning, task one-hot features, a SkillNorm-specific workstream.

All undetectable at the available sample size, or fixing a problem that did not exist: SkillNorm grade-4 recall@200 was already 1.000.

6 An openly reachable label channel

This section concerns a property of the challenge’s data hosting, and a negative methodological result about what a public leaderboard can tell you. It makes no claim about any participant, names none, and deliberately withholds the operational detail that would let a reader reconstruct a shortlist. It is included because the organisers asked for insights that would help a future edition.

6.1 The channel

The five graded WorkRB datasets are published on the Hugging Face Hub. Their `qrels/test` configurations were served, unauthenticated, through the public datasets-server API. A single request returns their sizes:

```
curl "https://datasets-server.huggingface.co/size?dataset=
TechWolf%2FSkill-extraction-House-graded&config=qrels"
```

I have two point observations, not continuous coverage. On 2026-07-30, five days before the test phase closed, all five graded task repositories exposed public `qrels/test` and `queries/test` splits whose counts matched the competition’s exactly. On 2026-08-04T01:08:39Z, an hour after the phase closed, I re-pulled and recorded the sizes: 528, 583, 582, 1265 and 450 `qrels` rows against 261, 338, 324, 500 and 450 test queries for House, Tech, TechWolf, SkillScape and SkillNorm respectively. I did not monitor the endpoint continuously, so I can say the splits were reachable during the competition and remained so after it, but not that they were reachable throughout.

The datasets’ own cards document the situation candidly. The published test labels are described as “real but binary” and “derived from the public non-graded ground truths”, with the 0–4 graded annotations “withheld during the ongoing RecSys-HR challenge”. So the withheld object (the graded annotation) was genuinely withheld. What remained reachable was a binary relevance judgement over the exact test queries.

The competition rules prohibit enriching training with test-source data, so this is a rules question rather than a grader vulnerability. I reported it to the organisers before results and the summary paper were finalised, and asked explicitly that my own submissions be included in whatever check followed.

The lesson for a future edition. A test set is not withheld because the interesting column has been dropped from a published table; what leaks is decided by what is reachable, not by what was intended to be released [58]. It is withheld when the rows themselves are unreachable. Gating the `qrels` splits for the duration of a competition costs nothing and removes the question entirely.

6.2 A negative result on leaderboard forensics

The natural follow-up question, whether a public leaderboard can reveal that such a channel was used, has a more interesting answer than I expected, and most of what I initially believed about it was wrong. Since the wrong beliefs are the transferable part, they are what this subsection reports.

The statistic that fails. The intuitive approach is *cross-sectional*: for each leaderboard row, compute the mean extraction score minus the normalisation score, and treat a large positive gap as anomalous. The reasoning is that a label source covering the extraction tasks would lift them and leave normalisation behind. It produces one number per row, it needs nothing but the public board, and on a short list of rows it appears to separate them cleanly.

It does not work, and I withdrew it. Computed over every row of the final board, a *majority* (57%) have a positive gap, so the statistic describes most of the field rather than separating a minority out of it. Worse for its intended use, its board maximum belongs to a row that was not on the shortlist I had built, and exceeds every row that was. A statistic whose extreme value ranks an unsuspected row above every suspected one is measuring something other than what I wanted it to measure: in this case, chiefly that extraction and normalisation are different problems with different attainable ceilings, and that most systems are better at the first. My

own entry sits on the other side of the same axis, with a *negative* gap, for the entirely mundane reason described in §5.

The statistic that is merely suggestive. The better-posed version is a *within-account step*: between consecutive submissions by the same account, does the subset of tasks a source would help move together and sharply while the rest stays flat? This at least controls for the system, which the cross-sectional version does not.

It is still weak, for a reason worth recording. A task column that does not move between two submissions most often means nothing more than that a per-task prediction file was reused unchanged, a routine thing to do when iterating on one task. On this board, frozen columns of that kind occur in roughly one consecutive pair in seven, across accounts spanning the full range of scores, including on submissions whose own descriptions identify them as reruns of the same baseline. A pattern with that base rate cannot carry much evidential weight on its own.

I also want to be accurate about its epistemic status, because I initially described it to myself as pre-registered and it is not quite that. The criterion was fixed before the scan was run, and the scan covered every account and every consecutive pair with none chosen in advance. But its thresholds were chosen to formalise a shape I had already noticed. That makes it a pre-registered *test* of a post-hoc *hypothesis*: better than choosing thresholds after seeing which accounts they flagged, and considerably weaker than a genuine prior prediction.

I am deliberately not publishing the thresholds. The leaderboard is public and the statistic is deterministic, so exact values would hand any reader a reproducible shortlist of accounts. That is precisely the accusation this paper declines to make, merely routed through the reader.

Where this ends up. I want to state my actual conclusion rather than leave it implied, because I reached the opposite one first and held it for some days:

I do not think any pattern in the public scores can distinguish a compliant system from a non-compliant one here. I was wrong to think otherwise.

Two things produced that reversal, and both are cheap enough that anyone attempting this analysis should run them first.

The first is the *base-rate check*: how often does the pattern fire across the whole population, rather than across the short list you were already looking at? Both statistics above survive scrutiny on a handful of rows and die on the full board.

The second is more uncomfortable, and it is the one I would emphasise. My own entry was the clearest outlier on the cross-sectional statistic: all nine of my placed rows fall on the opposite side of it. Which means the reference class I was implicitly comparing against was one I had constructed by removing everyone whose system did not resemble mine. A statistic that flags “systems unlike mine” will always find some, and it will always feel like a discovery.

Stated generally: a *cross-sectional* spread across accounts measures how the tasks differ. A *within-account step* measures how one system changed, which is better, but is still confounded by the most ordinary act in competitive machine learning, which is reusing a file you did not change. Neither is evidence of anything about a person. Rules compliance is a question for organisers with access to submissions and code, which is where I left it.

Provenance. An earlier and considerably more accusatory version of this analysis was put through an adversarial review in which independent reviewers were instructed to refute each claim, with refutation as the default under uncertainty. Of 72 claims examined across five review lenses, 14 survived, a 19% survival rate.

The detail worth recording is that the cross-sectional gap statistic was *not* among the claims that review rejected. It survived. I withdrew it myself afterwards, when the base-rate check

confirmed its arithmetic and destroyed its interpretation. A claim can pass adversarial review and still be wrong, which is an argument for running the cheap population-level check in addition to the expensive argument-level one, not instead of it.

7 Discussion

7.1 What actually mattered

Ranked by contribution per unit of effort, the campaign’s levers invert almost exactly the order in which one would naturally attempt them.

1. **Cross-encoder reranking** ($\approx +0.138$ across four steps). The expensive part, and the part that carries most of the score. Two architecture families, two-stage listwise training, GPU-days on a cluster.
2. **Counting statistics** (+0.0331). A smoothed per-skill relevance prior computed from training-fold candidate pools. Four columns of arithmetic, no GPU.
3. **Learning to rank over cheap signals** (+0.0291). Twelve retrieval lanes, most individually poor, fused by a LambdaRank model. No GPU.
4. **Exact string matching in the right place** (+0.0152 locally). An `altLabel` override applied at write time rather than offered as a feature. No GPU.
5. **Everything in the extended phase** (+0.0011). Weeks of it.

The three no-GPU levers total about +0.077: roughly a third of the gain over the fused base, for a rounding error of the cost. That is not an argument against cross-encoders: +0.138 was decisive for this entry, and I know of no configuration of the cheap levers alone that came close to it. Whether the top of *this* leaderboard was reachable without them I cannot say, since I have no visibility into any other entry’s architecture beyond a one-line self-description.

It is an argument about *sequencing*, and the criticism lands on me. The cheap levers were available from day one, and I did not reach the prior until the campaign was two-thirds gone. They also had the largest effects relative to their measurement error, which made them the ones a 286-query validation set could actually verify. By the time the campaign reached effects of +0.001, measurement had become more expensive than training.

7.2 Ensembles want decorrelation, not strength

The clearest technical lesson is the fate of Q8D. It was the strongest single reranker ever measured here, and every ensemble containing it failed, because the pool already held two Qwen-family members and Q8D’s marginal contribution was absorbed. The member that did pass, C4, scored 0.8245 alone. That put it above both DeBERTa columns already in the pool and below all four Qwen-family columns (Q at 0.8334, Q8 at 0.8358, Q2 at 0.8375, Q8D at 0.8407). It entered by *replacing* C3 rather than lengthening the ensemble.

But decorrelation is necessary, not sufficient, and the campaign measured the counterexample too. ELECTRA-large was the most lineage-decorrelated candidate available, sharing no pre-training with either family, and it failed in all three configurations it was tried in, including one at -0.0021 with zero seeds won. A member has to be both different *and* good enough that its differences are signal. Q8D was good and not different; ELECTRA was different and not good enough.

One level down, among the retrieval lanes, the description-scoring lanes dominate feature importance while being among the weakest standalone rankers. Across both levels, standalone quality is a poor guide on its own, but so is novelty on its own.

This is not a new observation in the ensembling literature, but it is worth restating in a competition context, where the instinct under time pressure is always to train a bigger version

of whatever currently scores highest.

7.3 On not submitting things

Two decisions in this campaign were about restraint rather than method.

The first: a deterministic per-task recombination of my own already-scored leaderboard rows was built, verified, and worth +0.0002. It was not submitted. Two reviews were run on it and they split. A technical review passed it. An honesty-and-optics review refused it on two grounds. The first was an ambiguity in the rules-letter: the rules require the submission file to be produced by the real benchmark tasks, and a recombination is assembled from files that were. The second was that +0.0002 changed no ranking, the nearest rival being 0.0072 below. Both reviews are on record. I report the split rather than claiming the call was obvious.

The second: submission quota is a genuine resource and should be modelled as one. Ten submissions, each returning one scalar hours later, is a bandit problem with an expensive arm. Two of our seven offset calibration points were obtained *for free*, by locally scoring two zero-shot baselines that already appeared on the public leaderboard. That trick (calibrate against someone else’s published number rather than spending your own) generalises to any competition with a public baseline row, and it is available on day one.

Its limit is worth stating alongside it. Those two free points read -0.0124 and -0.0139 , while the mature pipeline’s offset settled near -0.020 . A zero-shot baseline transfers better than a fitted system does, so the free estimate is optimistic and must be superseded once real rows exist. It is a starting prior, not a substitute for your own measurements.

7.4 Limitations

The system is expensive: the reranker ensemble requires multiple GPU-days and several large checkpoints at inference. Nothing here addresses latency, and the entire design assumes offline scoring of a fixed query set.

The validation set is small enough that even the protocol above cannot fully insure against selection effects. The pooled twenty-seed estimate of the final configuration (≈ 0.8644) sits below its discovery estimate (0.8647), in the direction the winner’s curse predicts, though a 0.0003 gap against a 0.0007 across-seed standard deviation is a single draw consistent with chance, so it illustrates the concern rather than measuring it.

TechWolf was never locally measurable, so one fifth of the final score was governed by a generalisation assumption rather than by evidence. The assumption held: TechWolf scored 0.8347, in line with the other extraction tasks. That is a fact about this competition, not a validated method.

Finally, this is a single competition entry, and the measurement protocol is offered as a description of what worked under one specific set of constraints rather than as a validated methodology.

8 Conclusion

The winning entry to the RecSys-HR 2026 WorkRB challenge is a conventional retrieve-then-rerank cascade: twelve fused retrieval lanes, an architecture-diverse ensemble of listwise-trained cross-encoders, and a LambdaRank fusion carrying a per-skill relevance prior and an exact-match override. It scored 0.8458 macro nDCG@100.

The components are not the interesting part. I make no claim about *why* this entry finished first: on score it did not separate itself from anything, standing tenth of twenty-nine on the public board at close, and the standing changed only after the organisers’ compliance review, a process I had no visibility into.

What I can claim is narrower and, I think, more useful. The decision procedure is what makes every number in this paper reportable: pre-register the configurations, compare them paired on identical folds, adopt only what clears a gate fixed in advance, confirm survivors on fresh seeds, and convert local scores into submission decisions through a measured and explicitly uncertain offset. In the final phase it adopted one configuration and rejected every other, including one that won ten seeds out of ten by a margin that missed the threshold in the fourth decimal place. Without it, the last three weeks would have been a search over noise, and I would have had no way to tell.

In a competition whose leading margins are smaller than its measurement error, the scarce resource is not compute. It is the ability to tell whether something worked.

Reproducibility and artefacts

The complete training code was submitted to the organisers as required by the challenge rules. The pipeline depends on the organisers’ WorkRB benchmark library and challenge repository, on ESCO v1.1, and on the publicly available checkpoints named in §3. Training data for stage 1 is `TechWolf/Synthetic-ESCO-skill-sentences` (CC-BY-4.0). Every quantitative claim in this paper traces to an entry in an append-only measurement ledger kept throughout the campaign or to a leaderboard record identified by submission number.

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Large language models were used throughout this project, principally for code and for drafting, including this paper. Every quantitative claim about the system traces to a recorded measurement or a leaderboard record rather than to a model’s summary of one.

Disclosure

The author competed solo as team `thylinao` and received prize money from the organisers for the first-place finish.

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A Selected hyperparameters

Component	Setting
Candidate pool	$0.6 z(\text{CurriculumMatch}) + 0.4 z(\text{ConTeXT})$, top-1000
Cross-encoder folds	3, query-grouped
Fusion folds	5, query-grouped, over 286 validation queries
Sequence budget	encoder members <code>max_length</code> 192 (whole pair) Qwen members 192 content tokens + 64-token template, 256 total
LoRA	$r = 16$, $\alpha = 32$, dropout 0.05, all attention and MLP projections
Learning rate (LoRA)	1×10^{-4} stage 1, 5×10^{-5} stage 2
Learning rate (full FT)	<code>bge-reranker-base</code> 2×10^{-5} / 1×10^{-5} <code>bge-reranker-v2-m3</code> 8×10^{-6} / 5×10^{-6} <code>deberta-v3-large</code> 1×10^{-5} / 6×10^{-6} (stage 1 / stage 2)
Optimiser	AdamW, weight decay 0, gradient clip 1.0
Schedule	linear warmup (50 steps stage 1, 20 stage 2; capped at <code>max_steps</code> /10) then linear decay to zero
Loss	LambdaLoss, NDCGLoss2++ scheme, $\mu = 10$, $\sigma = 1$
Stage-1 ladder	1 gold @4, 3 near @2, 5 mid @1, 6 deep @0 (15 docs/row)
LightGBM	300 trees, lr 0.05, 15 leaves, <code>min_child_samples</code> 50 subsample 0.8 (freq 1), <code>colsample_bytree</code> 0.8 $\alpha = 1.0$, $\lambda = 5.0$, <code>label_gain</code> = [0, 1, 3, 7, 15]
Bagging	10 seeds at final refit
Submission	top-500 targets per query, ID-keyed

B The final ensemble’s test members

Each cross-encoder contributes one or more test-side score matrices, combined by per-query z -score and then averaged before entering the fusion.

Member	Test matrices contributed
C2	full fit + folds 0, 1, 2
C4	full fit only
Q2	full fit + folds 0, 1, 2
Q8	full fit + fold 0

The asymmetry is an artefact of the deadline rather than a design choice: C4’s fold members had not finished training when the submission was built, and only Q8’s fold 0 was complete across all five tasks. A fold is listable only if it exists for every task, since the builder loads every listed member for every task.

Round	Configuration	10-seed	Δ paired	Wins	Verdict
<i>Baselines re-derived on the same seeds</i>					
–	v10cfg (A4,B2,C2,C3,Q2,Q8)	0.8641	–	–	incumbent
–	v11cfg lean (C2,C3,Q2,Q8)	0.8637	–	–	incumbent
–	incumbent blend	0.8650	–	–	incumbent
<i>Blend round 2: 5 configurations, 26 subsets</i>					
Blend	best subsets b0124 / b014	–	+0.0004	6–7/10	FAIL
Blend	Q-trio (Q, Q2, Q8) alone	0.8617	–	–	FAIL
Blend	no-DeBERTa (A4,B2,Q2,Q8)	0.8628	–	–	FAIL
Blend	no-Qwen (A4,B2,C2,C3)	0.8465	–	–	FAIL
<i>B-family</i>					
B-fam	A4,B,B2,C2,C3,Q2,Q8	0.8639	–0.0002	3/10	FAIL
B-fam	B,C2,C3,Q2,Q8	0.8638	–0.0003	4/10	FAIL
<i>New feature blocks</i>					
Feat	CE-dispersion column	0.8641	–0.0000	4/10	FAIL
Feat	ESCO attribute join	0.8646	+0.0005	6/10	FAIL
<i>Dense-column rounds</i>					
Q8D	C2,C3,Q2,Q8D	0.8635	–0.0002	4/10	FAIL
Q8D	C2,C3,Q2,Q8,Q8D	0.8642	+0.0005	8/10	FAIL(mean)
Q8D	A4,B2,C2,C3,Q2,Q8D	0.8636	–0.0005	3/10	FAIL
E	C2,C3,Q2,Q8,E	0.8637	+0.0001	4/10	FAIL
E	7-column + E	0.8637	–0.0004	3/10	FAIL
E	E,Q2,Q8	0.8615	–0.0021	0/10	FAIL
<i>The C4 round</i>					
C4	c4a: C2,C4,Q2,Q8	0.8647	+0.0011	10/10	PASS
C4	c4b: C2,C3,C4,Q2,Q8	0.8646	+0.0010	10/10	FAIL(strict >)
C4	c4c: A4,B2,C2,C4,Q2,Q8	0.8646	+0.0005	7/10	FAIL
<i>Round Q2D and the combined swap</i>					
Q2D	q2da	–	+0.0009	8/10	FAIL
Q2D	q2db	–	+0.0009	8/10	FAIL
Q2D	q2dc	–	+0.0006	9/10	FAIL
Comb	C2,C4,Q2D,Q8D	0.8639	+0.0002	5/10	FAIL
Add	lean + ESCO attributes	0.8636	–0.0001	4/10	FAIL

Table 4: The pre-registered ablation ledger of the extended phase: every registered round, with one pass. Standard deviation across seeds was within the 0.0015 gate for every row where it is reported; **c4a** measured 0.0007. **c4a** then passed fresh-seed confirmation on seeds 10–19 (+0.0010, 9/10) and was submitted. Note the three Round-Q2D near-misses at +0.0009 and +0.0006 with 8–9 seeds won: these are exactly the results a permissive protocol adopts and a fixed gate does not.